Ethan Xujia Fan

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Summary

Agentic AI | Software Development Life Cycle | Deep Learning | Reinforcement Learning | AI/Data Governance

Education

University of Ottawa, Ottawa, Master of Electrical and Computer Engineering, co-op — Jan. 2026 — Dec. 2027
Relevant Coursework: Deep Learning & Reinforcement Learning, Uncertainty Evaluation in ML

Carleton University, Ottawa, Honours Bachelor of Computer Science — Sept. 2020 — Dec. 2025

Technical Skills

AI/ML: Deep Learning, Reinforcement Learning, Agentic AI(RAG, MCP), AI Governance, Data Analysis
Programming: Java, Python, C, C++, SQL, React, Js
Operating Systems & Platforms: Linux, Shell, Colab, QNX rtos, Momentics IDE
Automation & Tools: Git, JIRA, TestRail, Agile/Scrum/SDLC

Work Experience

Neusoft (Amazon Vendor), Dalian, China — Jun. 2024 — Sept. 2024
Test Engineer

- Performed OTA and system-level testing (FSST, MSQ) for Amazon Fire OS TV; built Python automation pipelines in Linux and analyzed logs to diagnose failures
- Utilized JIRA and TestRail to manage test cycles; conducted automation testing with AWS KATS and DOTS, identifying scheduling bottlenecks and improving task isolation and execution efficiency
- Performed low-level Android system testing by modifying bootloaders and managing cache states
- Contributed to Fire OS 8 test planning and delivered presentations to Amazon developers and QA on framework usability, performance bottlenecks, and optimization suggestion

Projects

Enterprise Agentic AI Governance Platform - Sponsored by Kinaxis

- Designing a multi-tenant AI agent platform with policy enforcement, model routing, tenant-isolation, and audit logging

Neural Network Implementation & Performance Analysis (JAX vs PyTorch)

- Built and benchmarked JAX (grad, jit) and PyTorch neural networks on MNIST, analyzing scalability and optimization
- Evaluated execution paradigms and model reliability via uncertainty analysis, including calibration and robustness

Reinforcement Learning Stock Trading Agent

- Developed a trading agent for buy/sell/hold decisions using A2C, A3C, DQN, and PPO in the AnyTrading environment
- Optimized exploration strategies (epsilon-greedy, Boltzmann); identified DQN as best-performing model
- Co-authored a 9-page NeurIPS-style research paper detailing methodology and empirical results

Simulated Automated External Defibrillator in C++

- Designed and implemented an AED software simulation in C++ (Qt/Linux), modeling self-test, rhythm analysis, shock delivery, and post-shock monitoring within an SDLC and Agile workflow
- Followed an SDLC-based workflow with iterative development cycles aligned with Agile principles
- Produced UML and traceability artifacts, and conducted unit, integration, and system testing under varied operating conditions to validate functional reliability

Computer and Internet Security Study

- Studied cryptography, software vulnerabilities(RSA, TLS, Diffie-Hellman), Internet security, programming language vulnerabilities, and real-world attacks; explored privacy technologies (Tor, VPNs) and GDPR

Teaching Assistant Experience

CEG4166 Real-time Operating System - University of Ottawa — Jan. 2026 — Now

- Run labs, graded assignments and exams, delivered feedback

SEG2105 Intro to Software Engineering - University of Ottawa — Sept. 2025 — Dec. 2025

- Java, OOP, UML, Android, SQLite, and software testing
- Graded assignments/labs/exams, delivered feedback, and held weekly office hours
- Collaborated with the instructor and fellow TAs to align tutorial and lab content and course setting

COMP 1805 Discrete Structures I - Carleton University — May 2023 - Jul 2023 — May 2021 - Jul 2021

- Propositional logic, set theory, algorithm complexity, mathematical proof, and graph theory
- Conducted weekly office hours and graded 600+ assignments and exams, providing detailed feedback.